

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 3

Calculus - Differentiation & Applications

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find $\frac{dy}{dx}$ if $y = 3x^2 + 5x - 2$ [2 marks]

2. Differentiate: $y = (2x + 1)^3$ [3 marks]

3. Find the gradient of $y = x^2$ at $x = 3$ [2 marks]

4. If $y = \sqrt{x}$, find $\frac{dy}{dx}$ [2 marks]

5. Find $\frac{d}{dx}(\sin x + \cos x)$ [2 marks]

6. If $f(x) = 1/x$, find $f'(x)$ [2 marks]

7. Find the stationary point of $y = x^2 - 4x + 3$ [3 marks]

8. If $y = e^{\sin x}$, find $\frac{dy}{dx}$ [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Find the equation of tangent to $y = x^2 - 2x$ at point (3, 3). (5 marks)

10. A curve is $y = x^3 - 3x^2 + 2$. Find all stationary points and determine their nature. (6 marks)

11. Differentiate: $y = (x + 1)/(x - 1)$. (4 marks)

12. A particle moves such that $s = t^3 - 6t^2 + 9t$. Find velocity and acceleration at $t = 2$. (5 marks)

13. Find the rate of change of area of circle with radius 2 cm, if radius increases at 0.5 cm/s. (5 marks)

14. Sketch $y = x^3 - 3x$, showing all key features and stationary points. (6 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Differentiate $3x^2 + 5x - 2$

$$dy/dx = 6x + 5$$

Q2: Differentiate $(2x + 1)^3$

$$\text{Using chain rule: } dy/dx = 3(2x + 1)^2 \times 2 = 6(2x + 1)^2$$

Q3: Gradient at $x = 3$

$$dy/dx = 2x, \text{ at } x = 3: dy/dx = 6$$

Q4: Differentiate $\sqrt{x}$

$$y = x^{(1/2)}, dy/dx = (1/2)x^{(-1/2)} = 1/(2\sqrt{x})$$

Q5: Differentiate $\sin x + \cos x$

$$d/dx = \cos x - \sin x$$

Q6: Differentiate $1/x$

$$y = x^{(-1)}, f'(x) = -x^{(-2)} = -1/x^2$$

Q7: Stationary point

$$dy/dx = 2x - 4 = 0, x = 2, y = 4 - 8 + 3 = -1, \text{ point } (2, -1)$$

Q8: Differentiate e^x

$$dy/dx = e^x$$

SECTION B SOLUTIONS

Q9: Tangent equation

$$\text{At } (3,3): dy/dx = 2(3) - 2 = 4, y - 3 = 4(x - 3), y = 4x - 9$$

Q10: Stationary points of $y = x^3 - 3x^2 + 2$

$$dy/dx = 3x^2 - 6x = 3x(x - 2) = 0, x = 0 \text{ or } 2, \text{ At } x=0: y=2, (0,2) \text{ is local max; At } x=2: y=8-12+2=-2, (2,-2) \text{ is local min}$$

Q11: Differentiate $(x+1)/(x-1)$

$$\text{Using quotient rule: } dy/dx = [(x-1) - (x+1)]/(x-1)^2 = -2/(x-1)^2$$

Q12: Velocity and acceleration at t=2

$v = ds/dt = 3t^2 - 12t + 9$, $a = dv/dt = 6t - 12$, At $t=2$: $v = 12 - 24 + 9 = -3$ m/s, $a = 12 - 12 = 0$ m/s²

Q13: Rate of change of area

$A = \pi r^2$, $dA/dt = 2\pi r(dr/dt) = 2\pi(2)(0.5) = 2\pi$ cm²/s

Q14: Sketch curve

$y = x(x^2 - 3)$, stationary points at $x = \pm 1$, local max at $x = -1$, local min at $x = 1$